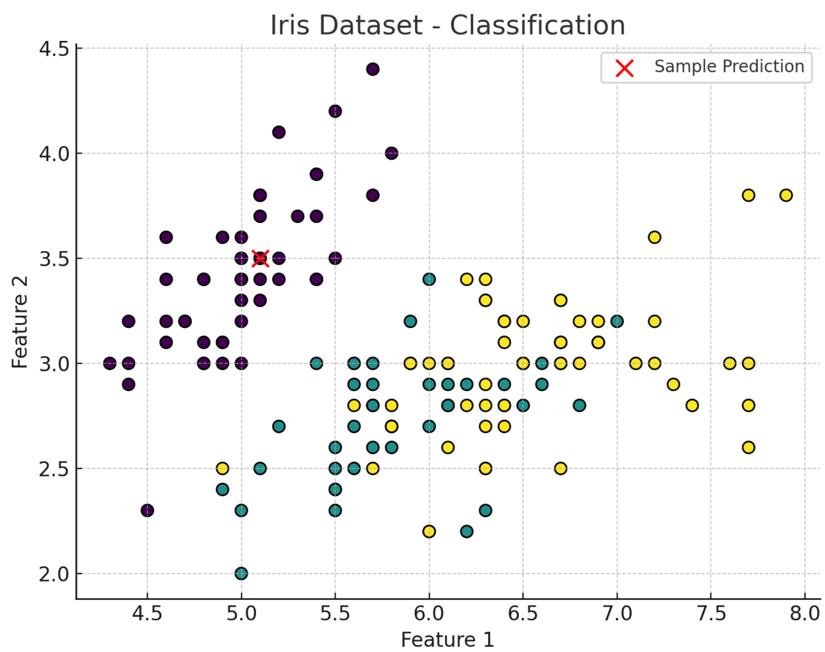
Hands-on Demo: Using Python and scikit-learn for classification:




```
from sklearn.datasets import load_iris
from sklearn.neighbors import KNeighborsClassifier

iris = load_iris()
X, y = iris.data, iris.target
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X, y)

sample = X[0]
pred_label = model.predict([sample])[0]
print("Predicted species:", iris.target_names[pred_label])
```

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2. Unsupervised Learning

Unsupervised learning uses unlabeled data, aiming to discover hidden patterns or groupings without explicit guidance. The goal is to infer the hidden structure from unlabeled data. There is no explicit “right answer” given; instead, the goal might be to discover hidden groupings, correlations, or anomalies. Unsupervised learning is often used for exploratory data analysis or as a preprocessing step.